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Deep Learning for Coders with fastai & PyTorch

AI Applications Without a PhD

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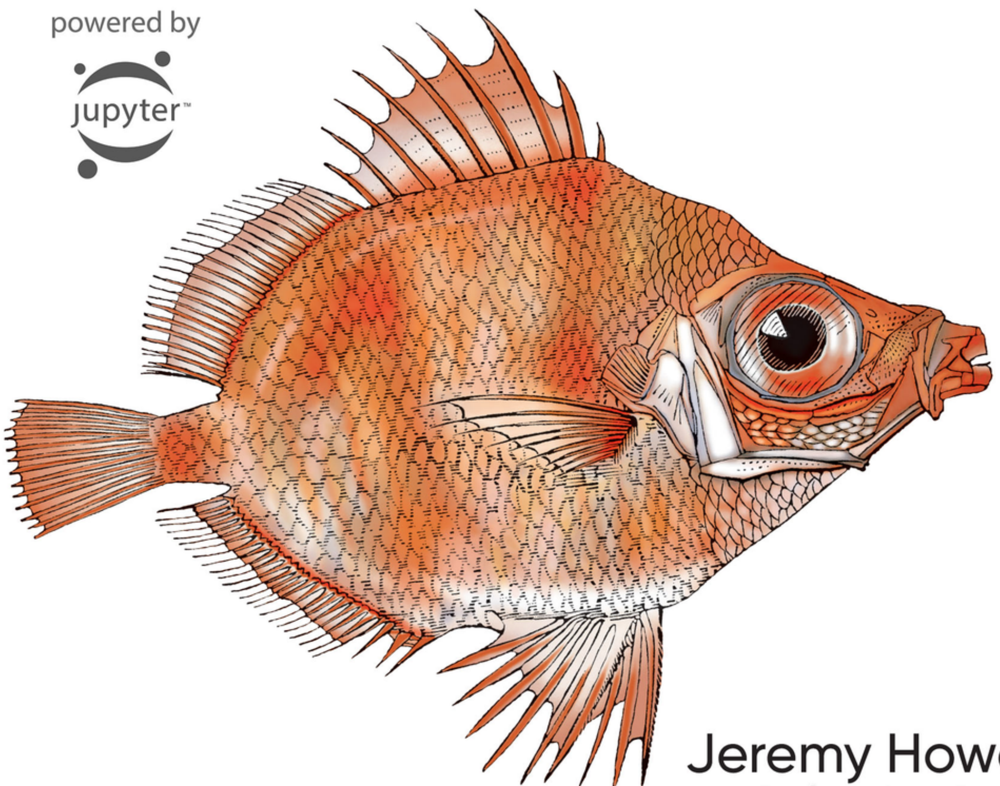

Jeremy Howard &
Sylvain Gugger

Foreword by Soumith Chintala

Praise for *Deep Learning for Coders with fastai and PyTorch*

If you are looking for a guide that starts at the ground floor and takes you to the cutting edge of research, this is the book for you. Don't let those PhDs have all the fun—you too can use deep learning to solve practical problems.

—Hal Varian, Emeritus Professor, UC Berkeley; Chief Economist, Google

As artificial intelligence has moved into the era of deep learning, it behooves all of us to learn as much as possible about how it works. Deep Learning for Coders provides a terrific way to initiate that, even for the uninitiated, achieving the feat of simplifying what most of us would consider highly complex.

—Eric Topol, Author, *Deep Medicine*; Professor, Scripps Research

Jeremy and Sylvain take you on an interactive—in the most literal sense as each line of code can be run in a notebook—journey through the loss valleys and performance peaks of deep learning. Peppered with thoughtful anecdotes and practical intuitions from years of developing and teaching machine learning, the book strikes the rare balance of communicating deeply technical concepts in a conversational and light-hearted way. In a faithful translation of fast.ai's award-winning online teaching philosophy, the book provides you with state-of-the-art practical tools and the real-world examples to put them to use. Whether you're a beginner or a veteran, this book will fast-track your deep learning journey and take you to new heights—and depths.

—Sebastian Ruder, Research Scientist, Deepmind

Jeremy Howard and Sylvain Gugger have authored a bravura of a book that successfully bridges the AI domain with the rest of the world. This work is a singularly substantive and insightful yet absolutely relatable primer on deep learning for anyone who is interested in this domain: a lodestar book amongst many in this genre.

—Anthony Chang, Chief Intelligence and Innovation Officer, Children's Hospital of Orange County

How can I "get" deep learning without getting bogged down? How can I quickly learn the concepts, craft, and tricks-of-the-trade using examples and code? Right here. Don't miss the new locus classicus for hands-on deep learning.

—Oren Etzioni, Professor, University of Washington; CEO, Allen Institute for AI

This book is a rare gem—the product of carefully crafted and highly effective teaching, iterated and refined over several years resulting in thousands of happy students. I'm one of them. fast.ai changed my life in a wonderful way, and I'm convinced that they can do the same for you.

—Jason Antic, Creator, DeOldify

Deep Learning for Coders is an incredible resource. The book wastes no time and teaches how to use deep learning effectively in the first few chapters. It then covers the inner workings of ML models and frameworks in a thorough but accessible fashion, which will allow you to understand and build upon them. I wish there was a book like this when I started learning ML, it is an instant classic!

—Emmanuel Ameisen, Author, *Building Machine Learning Powered Applications*

"Deep Learning is for everyone," as we see in Chapter 1, Section 1 of this book, and while other books may make similar claims, this book delivers on the claim. The authors have extensive knowledge of the field but are able to describe it in a way that is perfectly suited for a reader with experience in programming but not in machine learning. The book shows examples first, and only covers theory in the context of concrete examples. For most people, this is the best way to learn.The book does an impressive job of covering the key applications of deep learning in computer vision, natural language processing, and tabular data processing, but also covers key topics like data ethics that some other books miss. Altogether, this is one of the best sources for a programmer to become proficient in deep learning.

—Peter Norvig, Director of Research, Google

Gugger and Howard have created an ideal resource for anyone who has ever done even a little bit of coding. This book, and the fast.ai courses that go with it, simply and practically demystify deep learning using a hands-on approach, with pre-written code that you can explore and re-use. No more slogging through theorems and proofs about abstract concepts. In Chapter 1 you will build your first deep learning model, and by the end of the book you will know how to read and understand the Methods section of any deep learning paper.

—Curtis Langlotz, Director, Center for Artificial Intelligence in Medicine and Imaging, Stanford University

This book demystifies the blackest of black boxes: deep learning. It enables quick code experimentations with a complete python notebook. It also dives into the ethical implication of artificial intelligence, and shows how to avoid it from becoming dystopian.

—Guillaume Chaslot, Fellow, Mozilla

As a pianist turned OpenAI researcher, I'm often asked for advice on getting into Deep Learning, and I always point to fastai. This book manages the seemingly impossible—it's a friendly guide to a complicated subject, and yet it's full of cutting-edge gems that even advanced practitioners will love.

—Christine Payne, Researcher, OpenAI

An extremely hands-on, accessible book to help anyone quickly get started on their deep learning project. It's a very clear, easy to follow and honest guide to practical deep learning. Helpful for beginners to executives/managers alike. The guide I wished I had years ago!

—Carol Reiley, Founding President and Chair, Drive.ai

Jeremy and Sylvain's expertise in deep learning, their practical approach to ML, and their many valuable open-source contributions have made then key figures in the PyTorch community. This book, which continues the work that they and the fast.ai community are doing to make ML more accessible, will greatly benefit the entire field of AI.

—Jerome Pesenti, Vice President of AI, Facebook

Deep Learning is one of the most important technologies now, responsible for many amazing recent advances in AI. It used to be only for PhDs, but no longer! This book, based on a very popular fast.ai course, makes DL accessible to anyone with programming experience. This book teaches the "whole *game", with excellent hands-on examples and a companion interactive site. And PhDs will also learn a lot.*

—Gregory Piatetsky-Shapiro, President, KDnuggets

An extension of the fast.ai course that I have consistently recommended for years, this book by Jeremy and Sylvain, two of the best deep learning experts today, will take you from beginner to qualified practitioner in a matter of months. Finally, something positive has come out of 2020!

—Louis Monier, Founder, Altavista; former Head of Airbnb AI Lab

We recommend this book! Deep Learning for Coders with fastai and PyTorch uses advanced frameworks to move quickly through concrete, real-world artificial intelligence or automation tasks. This leaves time to cover usually neglected topics, like safely taking models to production and a much-needed chapter on data ethics.

—John Mount and Nina Zumel, Authors, *Practical Data Science with R*

This book is "for Coders" and does not require a PhD. Now, I do have a PhD and I am no coder, so why have I been asked to review this book? Well, to tell you how friggin awesome it really is!

Within a couple of pages from Chapter 1 you'll figure out how to get a state-of-the-art network able to classify cat vs. dogs in 4 lines of code and less than 1 minute of computation. Then you land Chapter 2, which takes you from model to production, showing how you can serve a webapp in no time, without any HTML or JavaScript, without owning a server.

I think of this book as an onion. A complete package that works using the best possible settings. Then, if some alterations are required, you can peel the outer layer. More tweaks? You can keep discarding shells. Even more? You can go as deep as using bare PyTorch. You'll have three independent voices accompanying you around your journey along this 600 page book, providing you guidance and individual perspective.

—Alfredo Canziani, Professor of Computer Science, NYU

Deep Learning for Coders with fastai and PyTorch is an approachable conversationally-driven book that uses the whole game approach to teaching deep learning concepts. The book focuses on getting your hands dirty right out of the gate with real examples and bringing the reader along with reference concepts only as needed. A practitioner may approach the world of deep learning in this book through hands-on examples in the first half, but will find themselves naturally introduced to deeper concepts as they traverse the back half of the book with no pernicious myths left unturned.

—Josh Patterson, Patterson Consulting

Jeremy, Sylvain, and Rachel are the absolute masters of creating accessible tools and building community around AI. This is yet another installment of the fast.ai team creating an amazing resource that will help onboard the next hundred thousand aspiring AI researchers globally. Congrats!!!

—Joe Spisak, PyTorch Product Manager, Facebook

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Jeremy Howard and Sylvain Gugger

Deep Learning for Coders with fastai and PyTorch

by Jeremy Howard and Sylvain Gugger

Printed in Canada.

Published by O’Reilly Media, Inc., 1005 Gravenstein Highway North, Sebastopol, CA 95472.

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Cover Designer: Karen Montgomery

Illustrator: Rebecca Demarest

July 2020: First Edition

Revision History for the First Edition

- 2020-06-29: First Release

- 2020-09-18: Second Release
- 2020-12-18: Third Release
- 2021-05-07: Fourth Release
- 2021-11-05: Fifth Release

See http://oreilly.com/catalog/errata.csp?isbn=9781492045526 for release details.

978-1-492-04552-6

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Preface

Deep learning is a powerful new technology, and we believe it should be applied across many disciplines. Domain experts are the most likely to find new applications of it, and we need more people from all backgrounds to get involved and start using it.

That's why Jeremy cofounded fast.ai, to make deep learning easier to use through free online courses and software. Sylvain is a research engineer at Hugging Face. Previously he was a research scientist at fast.ai and a former mathematics and computer science teacher in a program that prepares students for entry into France's elite universities. Together, we wrote this book in the hope of putting deep learning into the hands of as many people as possible.

Who This Book Is For

If you are a complete beginner to deep learning and machine learning, you are most welcome here. Our only expectation is that you already know how to code, preferably in Python.

NO EXPERIENCE? NO PROBLEM!

If you don't have any experience coding, that's OK too! The first three chapters have been explicitly written in a way that will allow executives, product managers, etc. to understand the most important things they'll need to know about deep learning. When you see bits of code in the text, try to look them over to get an intuitive sense of what they're doing. We'll explain them line by line. The details of the syntax are not nearly as important as a high-level understanding of what's going on.

If you are already a confident deep learning practitioner, you will also find a lot here. In this book, we will be showing you how to achieve world-class results, including techniques from the latest research. As we will show, this doesn't require advanced mathematical training or years of study. It just requires a bit of common sense and tenacity.

What You Need to Know

As we said before, the only prerequisite is that you know how to code (a year of experience is enough), preferably in Python, and that you have at least followed a high school math course. It doesn't matter if you remember little of it right now; we will brush up on it as needed. Khan Academy has great free resources online that can help.

We are not saying that deep learning doesn't use math beyond high school level, but we will teach you (or direct you to resources that will teach you) the basics you need as we cover the subjects that require them.

The book starts with the big picture and progressively digs beneath the surface, so you may need, from time to time, to put it aside and go learn some additional topic (a way of coding something or a bit of math). That is completely OK, and it's the way we intend the book to be read. Start browsing it, and consult additional resources only as needed.

Please note that Kindle or other ereader users may need to double-click images to view the full-sized versions.

ONLINE RESOURCES

All the code examples shown in this book are available online in the form of Jupyter notebooks (don't worry; you will learn all about what Jupyter is in Chapter 1). This is an interactive version of the book, where you can actually execute the code and experiment with it. See the book's website for more information. The website also contains up-to-date information on setting up the various tools we present and some additional bonus chapters.

What You Will Learn

After reading this book, you will know the following:

- How to train models that achieve state-of-the-art results in
  - Computer vision, including image classification (e.g., classifying pet photos by breed) and image localization and detection (e.g., finding the animals in an image)
  - Natural language processing (NLP), including document classification (e.g., movie review sentiment analysis) and language modeling
  - Tabular data (e.g., sales prediction) with categorical data, continuous data, and mixed data, including time series
  - Collaborative filtering (e.g., movie recommendation)
- How to turn your models into web applications
- Why and how deep learning models work, and how to use that knowledge to improve the accuracy, speed, and reliability of your models
- The latest deep learning techniques that really matter in practice
- How to read a deep learning research paper
- How to implement deep learning algorithms from scratch
- How to think about the ethical implications of your work, to help ensure that you're making the world a better place and that your work isn't misused for harm

See the table of contents for a complete list, but to give you a taste, here are some of the techniques covered (don't worry if none of these words mean anything to you yet—you'll learn them all soon):

- Affine functions and nonlinearities
- Parameters and activations
- Random initialization and transfer learning
- SGD, Momentum, Adam, and other optimizers
- Convolutions
- Batch normalization
- Dropout
- Data augmentation

- Weight decay
- ResNet and DenseNet architectures
- Image classification and regression
- Embeddings
- Recurrent neural networks (RNNs)
- Segmentation
- U-Net
- And much more!

> **CHAPTER QUESTIONNAIRES**
>
> If you look at the end of each chapter, you'll find a questionnaire. That's a great place to see what we cover in each chapter, since (we hope!) by the end of each one, you'll be able to answer all the questions there. In fact, one of our reviewers (thanks, Fred!) said that he likes to read the questionnaire *first*, before reading the chapter, so he knows what to look out for.

O'Reilly Online Learning

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Foreword

In a very short time, deep learning has become a widely useful technique, solving and automating problems in computer vision, robotics, healthcare, physics, biology, and beyond. One of the delightful things about deep learning is its relative simplicity. Powerful deep learning software has been built to make getting started fast and easy. In a few weeks, you can understand the basics and get comfortable with the techniques.

This opens up a world of creativity. You start applying it to problems that have data at hand, and you feel wonderful seeing a machine solving problems for you. However, you slowly feel yourself getting closer to a giant barrier. You built a deep learning model, but it doesn't work as well as you had hoped. This is when you enter the next stage, finding and reading state-of-the-art research on deep learning.

However, there's a voluminous body of knowledge on deep learning, with three decades of theory, techniques, and tooling behind it. As you read through some of this research, you realize that humans can explain simple things in really complicated ways. Scientists use words and mathematical notation in these papers that appear foreign, and no textbook or blog post seems to cover the necessary background that you need in accessible ways. Engineers and programmers assume you know how GPUs work and have knowledge about obscure tools.

This is when you wish you had a mentor or a friend that you could talk to. Someone who was in your shoes before, who knows the tooling and the math—someone who could guide you through the best research, state-of-the-art techniques, and advanced engineering, and make it comically simple. I was in your shoes a decade ago, when I was breaking into the field of machine learning. For years, I struggled to understand papers that had a little bit of math in them. I had good mentors around me, which helped me greatly, but it took me many years to get comfortable with machine learning and deep learning. That motivated me to coauthor PyTorch, a software framework to make deep learning accessible.

Jeremy Howard and Sylvain Gugger were also in your shoes. They wanted to learn and apply deep learning, without any previous formal training as ML scientists or engineers. Like me, Jeremy and Sylvain learned gradually over the years and eventually became experts and leaders. But unlike me, Jeremy and Sylvain selflessly put a huge amount of energy into making sure others don’t have to take the painful path that they took. They built a great course called fast.ai that makes cutting-edge deep learning techniques accessible to people who know basic programming. It has graduated hundreds of thousands of eager learners who have become great practitioners.

In this book, which is another tireless product, Jeremy and Sylvain have constructed a magical journey through deep learning. They use simple words and introduce every concept. They bring cutting-edge deep learning and state-of-the-art research to you, yet make it very accessible.

You are taken through the latest advances in computer vision, dive into natural language processing, and learn some foundational math in a 500-page delightful ride. And the ride doesn’t stop at fun, as they take you through shipping your ideas to production. You can treat the fast.ai community, thousands of practitioners online, as your extended family, where individuals like you are available to talk and ideate small and big solutions, whatever the problem may be.

I am very glad you’ve found this book, and I hope it inspires you to put deep learning to good use, regardless of the nature of the problem.

Soumith Chintala
Cocreator of PyTorch

Part I. Deep Learning in Practice

Chapter 1. Your Deep Learning Journey

Hello, and thank you for letting us join you on your deep learning journey, however far along that you may be! In this chapter, we will tell you a little bit more about what to expect in this book, introduce the key concepts behind deep learning, and train our first models on different tasks. It doesn’t matter if you don’t come from a technical or a mathematical background (though it’s OK if you do too!); we wrote this book to make deep learning accessible to as many people as possible.

Deep Learning Is for Everyone

A lot of people assume that you need all kinds of hard-to-find stuff to get great results with deep learning, but as you'll see in this book, those people are wrong. Table 1-1 lists a few things you *absolutely don't need* for world-class deep learning.

Table 1-1. What you don't need for deep learning

Myth (don't need)	Truth
Lots of math	High school math is sufficient.
Lots of data	We've seen record-breaking results with <50 items of data.
Lots of expensive computers	You can get what you need for state-of-the-art work for free.

Deep learning is a computer technique to extract and transform data—with use cases ranging from human speech recognition to animal imagery classification—by using multiple layers of neural networks. Each of these layers takes its inputs from previous layers and progressively refines them. The layers are trained by algorithms that minimize their errors and improve their accuracy. In this way, the network learns to perform a specified task. We will discuss training algorithms in detail in the next section.

Deep learning has power, flexibility, and simplicity. That's why we believe it should be applied across many disciplines. These include the social and physical sciences, the arts, medicine, finance, scientific research, and many more. To give a personal example, despite having no background in medicine, Jeremy started Enlitic, a company that uses deep learning algorithms to diagnose illness and disease. Within months of starting the company, it was announced that its algorithm could identify malignant tumors more accurately than radiologists.

Here's a list of some of the thousands of tasks in different areas for which deep learning, or methods heavily using deep learning, is now the best in the world:

Natural language processing (NLP)

Answering questions; speech recognition; summarizing documents; classifying documents; finding names, dates, etc. in documents; searching for articles mentioning a concept

Computer vision

Satellite and drone imagery interpretation (e.g., for disaster resilience), face recognition, image captioning, reading traffic signs, locating pedestrians and vehicles in autonomous vehicles

Medicine

Finding anomalies in radiology images, including CT, MRI, and X-ray images; counting features in pathology slides; measuring features in ultrasounds; diagnosing diabetic retinopathy

Biology

Folding proteins; classifying proteins; many genomics tasks, such as tumor-normal sequencing and classifying clinically actionable genetic mutations; cell classification; analyzing protein/protein interactions

Image generation

Colorizing images, increasing image resolution, removing noise from images, converting images to art in the style of famous artists

Recommendation systems

Web search, product recommendations, home page layout

Playing games

Chess, Go, most Atari video games, and many real-time strategy games

Robotics

Handling objects that are challenging to locate (e.g., transparent, shiny, lacking texture) or hard to pick up

Other applications

Financial and logistical forecasting, text to speech, and much, much more…

What is remarkable is that deep learning has such varied applications, yet nearly all of deep learning is based on a single innovative type of model: the neural network.

But neural networks are not, in fact, completely new. In order to have a wider perspective on the field, it is worth starting with a bit of history.

Neural Networks: A Brief History

In 1943 Warren McCulloch, a neurophysiologist, and Walter Pitts, a logician, teamed up to develop a mathematical model of an artificial neuron. In their paper "A Logical Calculus of the Ideas Immanent in Nervous Activity," they declared the following:

> *Because of the "all-or-none" character of nervous activity, neural events and the relations among them can be treated by means of propositional logic. It is found that the behavior of every net can be described in these terms.*

McCulloch and Pitts realized that a simplified model of a real neuron could be represented using simple addition and thresholding, as shown in Figure 1-1. Pitts was self-taught, and by age 12, had received an offer to study at Cambridge University with the great Bertrand Russell. He did not take up this invitation, and indeed throughout his life did not accept any offers of advanced degrees or positions of authority. Most of his famous work was done while he was homeless. Despite his lack of an officially recognized position and increasing social isolation, his work with McCulloch was influential and was taken up by a psychologist named Frank Rosenblatt.

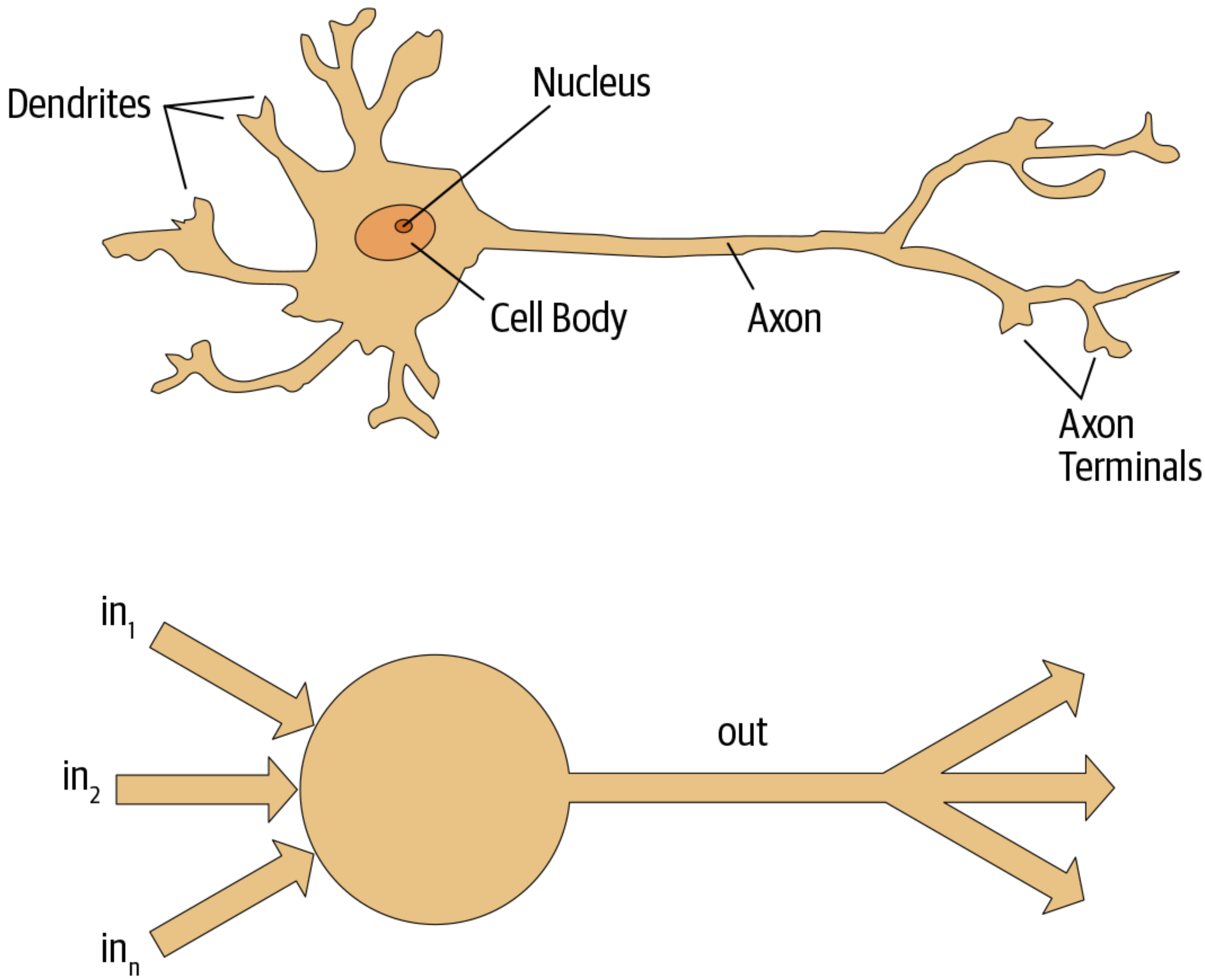

Figure 1-1. Natural and artificial neurons

Rosenblatt further developed the artificial neuron to give it the ability to learn. Even more importantly, he worked on building the first device that used these principles, the Mark I Perceptron. In “The Design of an Intelligent Automaton,” Rosenblatt wrote about this work: “We are now about to witness the birth of such a machine—a machine capable of perceiving, recognizing and identifying its surroundings without any human training or control.” The perceptron was built and was able to successfully recognize simple shapes.

An MIT professor named Marvin Minsky (who was a grade behind Rosenblatt at the same high school!), along with Seymour Papert, wrote a book called *Perceptrons* (MIT Press) about Rosenblatt’s invention. They showed that a single layer of these devices was unable to learn some simple but critical mathematical functions (such as XOR). In the same book, they also showed that using multiple layers of the devices would allow these limitations to be addressed. Unfortunately, only the first of these insights was widely recognized. As a result, the global academic community nearly entirely gave up on neural networks for the next two decades.

Perhaps the most pivotal work in neural networks in the last 50 years was the multi-volume *Parallel Distributed Processing* (PDP) by David Rumelhart, James McClelland, and the PDP Research Group, released in 1986 by MIT Press. Chapter 1 lays out a similar hope to that shown by Rosenblatt:

> *People are smarter than today's computers because the brain employs a basic computational architecture that is more suited to deal with a central aspect of the natural information processing tasks that people are so good at.…We will introduce a computational framework for modeling cognitive processes that seems…closer than other frameworks to the style of computation as it might be done by the brain.*

The premise that PDP is using here is that traditional computer programs work very differently from brains, and that might be why computer programs had been (at that point) so bad at doing things that brains find easy (such as recognizing objects in pictures). The authors claimed that the PDP approach was "closer than other frameworks" to how the brain works, and therefore it might be better able to handle these kinds of tasks.

In fact, the approach laid out in PDP is very similar to the approach used in today's neural networks. The book defined parallel distributed processing as requiring the following: